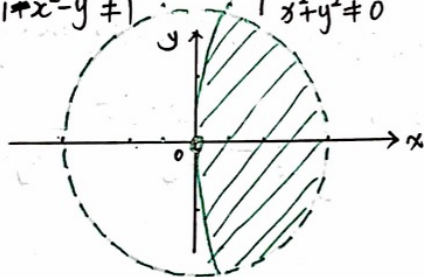


§ 9-1

1. 求函数 $z = \frac{\sqrt{4x-y^2}}{\ln(1-x^2-y^2)}$ 的定义域并画出定义域的图形.

解: $\begin{cases} 4x-y^2 \geq 0 \\ 1-x^2-y^2 > 0 \\ 1-x^2-y^2 \neq 1 \end{cases}$ 即 $\begin{cases} 4x \geq y^2 \\ x^2+y^2 < 1 \\ x^2+y^2 \neq 0 \end{cases}$ $\therefore$ 定义域为 $\{(x,y) \mid y^2 \leq 4x, 0 < x^2+y^2 < 1\}$.



2. 讨论函数 $f(x,y) = \begin{cases} \frac{x^2+xy}{x^2+y^2}, & x^2+y^2 \neq 0 \\ 0, & x^2+y^2 = 0 \end{cases}$ 的连续性.

解: $\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=kx}} \frac{x^2+xy}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{x^2+kx^2}{x^2+k^2x^2} = \frac{1+k}{1+k^2}$.

$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{x^2+xy}{x^2+y^2}$ 不存在.

因此, $f(x,y)$ 在 $(0,0)$ 处不连续.

§ 9-2

1. 若 $f(x,y)$ 在点 (x_0, y_0) 处的两个偏导数都存在, 则 (C).

(A) $f(x,y)$ 在点 (x_0, y_0) 的某个邻域内有界; 0 两码事

(B) $f(x,y)$ 在点 (x_0, y_0) 的某个邻域内连续; X

(C) $f(x,y)$ 在 x_0 处连续, $f(x_0, y)$ 在 y_0 处连续; ✓

(D) $f(x,y)$ 在点 (x_0, y_0) 处连续. X

2. 设 $f(x,y) = y(x-1)^2 + x(y-2)^2$, 则 $f_x(1,2) = 0$.

$$f_x(1,2) = [f(x,2)]'_x \Big|_{x=1} = [2(x-1)^2]'_x \Big|_{x=1} = 4(x-1) \Big|_{x=1} = 0$$

3. 设 $f(x, y) = \begin{cases} (x^2 + y) \sin \frac{1}{x^2 + y^2}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0. \end{cases}$ 求 $f_x(0, 0)$, $f_y(0, 0)$.

解: $f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x^2 \sin \frac{1}{\Delta x^2}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \Delta x \cdot \sin \frac{1}{\Delta x^2} = 0.$

(因为 $\Delta x \rightarrow 0$, 而 $|\sin \frac{1}{\Delta x^2}| \leq 1$. 无穷小乘以有界函数仍是无穷小量).

$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, 0 + \Delta y) - f(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{\Delta y \cdot \sin \frac{1}{\Delta y^2}}{\Delta y} = \lim_{\Delta y \rightarrow 0} \sin \frac{1}{\Delta y^2}$ 不存在.

4. 求下列函数的偏导数.

(1) $z = e^{\frac{y}{x}}$;

解: $\frac{\partial z}{\partial x} = e^{\frac{y}{x}} \cdot (-\frac{y}{x^2}) = -\frac{y}{x^2} e^{\frac{y}{x}}.$

$\frac{\partial z}{\partial y} = e^{\frac{y}{x}} \cdot (\frac{1}{x}) = \frac{1}{x} e^{\frac{y}{x}}.$

(2) $z = (1 + xy)^y$;

解: $\frac{\partial z}{\partial x} = y(1 + xy)^{y-1} \cdot y = y^2(1 + xy)^{y-1}.$

$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} (e^{y \ln(1+xy)}) = e^{y \ln(1+xy)} \cdot [\ln(1+xy) + y \cdot \frac{x}{1+xy}].$

$= (1 + xy)^y \cdot [\ln(1 + xy) + \frac{xy}{1 + xy}].$

5. 设 $z = x^y$, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

解: $\frac{\partial z}{\partial x} = y x^{y-1}$.

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} (y x^{y-1}) = x^{y-1} + y \cdot x^{y-1} \ln x = x^{y-1} (1 + y \ln x).$$

6. $f_{xy}(x, y)$ 与 $f_{yx}(x, y)$ 在区域 D 内连续是它们在 D 内相等的 (B).

(A) 必要条件; (B) 充分条件; (C) 充分必要条件 (D) 无关条件.

§ 9-3

1. 函数 $f(x, y)$ 在点 (x, y) 各一阶偏导数连续是 $f(x, y)$ 在该点可微分的 (B).

(A) 必要条件; (B) 充分条件; (C) 充分必要条件 (D) 无关条件.

2. 设函数 $u = x^{yz}$, 求 du .

解: $\frac{\partial u}{\partial x} = yz \cdot x^{yz-1}$;

$$\frac{\partial u}{\partial y} = x^{yz} (\ln x) \cdot z$$

$$\frac{\partial u}{\partial z} = x^{yz} (\ln x) y.$$

$$\begin{aligned} \therefore du &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ &= yz \cdot x^{yz-1} dx + z x^{yz} \ln x dy + y x^{yz} \ln x dz. \end{aligned}$$